

NVIDIA (NVDA) — Deep-Tech Investment Analysis

Technologist-first, dual-sided write-up. Maps 1:1 to `DECK_BLUEPRINT.md`.

In-session analysis — no API key. Produced by the Claude analyst desk reading `targets/NVDA/_brief.txt`. Data as-of **2026-06-22**; spot **\$209.91** [FACT]. Every load-bearing number is tagged [FACT] / [ESTIMATE] / [ASSUMPTION] / [ATTRIBUTED]. No figures are invented — canonical numbers only. Where the bull and bear disagree, both readings are shown and reconciled by the senior call.

The one-line hook (Slide 1): *NVDA moved the unit of sale from a chip to a coherent rack — and the whole investment case now collapses onto a single technical number: the accelerator dollar-share it retains as inference migrates to custom ASICs.*

Senior call up front: LONG-BIASED but two-sided, with entry discipline — accumulate below ~\$195, do not chase at \$210. At ~75% retained share and a \$440B 2029 TAM the stock is roughly fairly valued (~+5% IRR at spot); the edge is in the *entry*, not the thesis. Flip to a paired underweight only on a confirmed share-loss tell, never on a margin headline alone.

Q1 — The Technical "Moat" (Blueprint slides 6–9, 24)

ELI5 (slide 6): what NVDA actually sells now

Picture a giant AI model as a single huge brain. One chip can only hold a slice of it, so the slices have to constantly talk to each other. **NVDA's trick is not making the fastest single chip — it is wiring 72 chips together so tightly they behave like one gigantic chip.** Imagine 72 office workers who, instead of emailing, share one desk and one whiteboard at conversational speed. That shared whiteboard is the moat. A rival can build a faster single worker (a custom ASIC), but it still can't replicate the 72-at-one-desk room — *and* it can't replace the 18 years of software everyone already wrote for NVDA's room.

The specific engineering choice that is the moat (slide 9)

The defensible advantage is **not any single die** — a hyperscaler ASIC can match TOPS/\$ on one piece of silicon. It is the **scale-up coherence domain**:

- **Interconnect:** NVLink/NVSwitch fuse 72 GPUs into one **~130 TB/s, ~1.8 TB/s/GPU** non-blocking fabric (**>14x PCIe Gen5**), so a trillion-parameter Mixture-of-Experts model behaves as a single accelerator with most expert-hops staying in-rack [ATTRIBUTED]. SemiAnalysis measured GB200 NVL72 at **up to ~28x MI355X throughput on DeepSeek-R1 MoE** [ATTRIBUTED].
- **Software:** ~18 years of CUDA (cuDNN / TensorRT-LLM / NCCL / Triton) makes the switching cost *the entire ML software stack* [ATTRIBUTED].
- **Packaging allocation:** CoWoS-L 2.5D packaging (LSI bridges, ~5.5–6x reticle area) that NVDA **pre-books at TSMC** [ATTRIBUTED].
- **Cadence:** an annual co-design cycle (Hopper→Blackwell→Rubin→Feynman) re-touches the perf/watt frontier every ~12 months — faster than any challenger's design loop [ATTRIBUTED].

Three independent moats — software, interconnect, packaging allocation — sit on top of each other; a competitor must beat all three at once.

The honest counter (the technologist's caveat)

The moat's load-bearing claim — *"ASICs erode inference but NOT frontier-training / large-MoE economics"* — is the one place the adversarial verification dented the bull (LONG LB1: **weakened** on all three lenses). Real-world frontier behavior contradicts the neat line: **Anthropic committed (Nov-2025)**

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toward ~1M Google TPUs by 2027; Meta is reportedly negotiating TPU leasing from 2026; Gemini trains frontier/MoE entirely on TPU [ATTRIBUTED]. So the moat is real at the *inference* margin and at the *frontier core* — but the frontier is no longer 100% NVDA-only. That is exactly why the whole case reduces to *retained share* rather than "the moat is binary."

Q2 — Commercial Scaling: specs → profit / margin / share (Blueprint slides 27, 20–22)

The proof chain — the engineering choice monetizes at a level nothing else in semis approaches. FY26 (anchor, all [FACT]):

Metric	FY26	Tag
Revenue	215.94B (+65.5% YoY)	[FACT]
Gross margin	71.1% (peaked 75.0% FY25)	[FACT]
Operating margin	60.4%	[FACT]
Net margin	55.6%	[FACT]
Net income	120.07B	[FACT]
Operating cash flow	102.72B	[FACT]
R&D	18.50B (8.6% of sales)	[FACT]
Cash / LT debt / equity	10.61B / 7.47B / 157.29B	[FACT]

That is a **hardware bill-of-materials earning software economics**. The nearest custom-silicon enabler **AVGO runs ~40% op margin; AMD ~11%** [FACT]. R&D at only 8.6% of sales means the annual cadence **self-funds**, and a near-unlevered balance sheet means the OpenAI/neocloud investments are paid from cash flow, not debt — categorically unlike the Lucent/Nortel vendor-financing template the bears invoke.

Share-economics caveat (slide 27, honest version): the verification *weakened but did not refute* the inference-leakage thesis (SHORT LB2: **weakened**, market lens **high**). ASIC units grew **+44.6% vs merchant GPU +16.1% in 2026** [ATTRIBUTED]; bears model NVDA inference share falling from ~90% toward **20–30% by 2028** [ATTRIBUTED]. The cap on the threat: much of the ASIC "40–65% lower TCO" [ATTRIBUTED] is hyperscalers pricing captive silicon at *cost* (no ~71% NVDA margin embedded), and only ~4–5 buyers can build captive ASICs at all. **Blended ASP/GM dilutes if inference share leaks — even if unit count holds**. That is the single lever this whole report turns on.

Q3 — PxQ + how the end market grows (Blueprint slides 20–22, 10)

PxQ redefined upward. NVDA shifted the sellable SKU from a ~\$30–35K discrete GPU to a ~\$3M GB200 NVL72 rack, with Vera Rubin NVL72 reportedly up to ~\$8.8M [ATTRIBUTED] — capturing the tray, NVLink switch, Grace CPU and networking margin a chip-only vendor never sees.

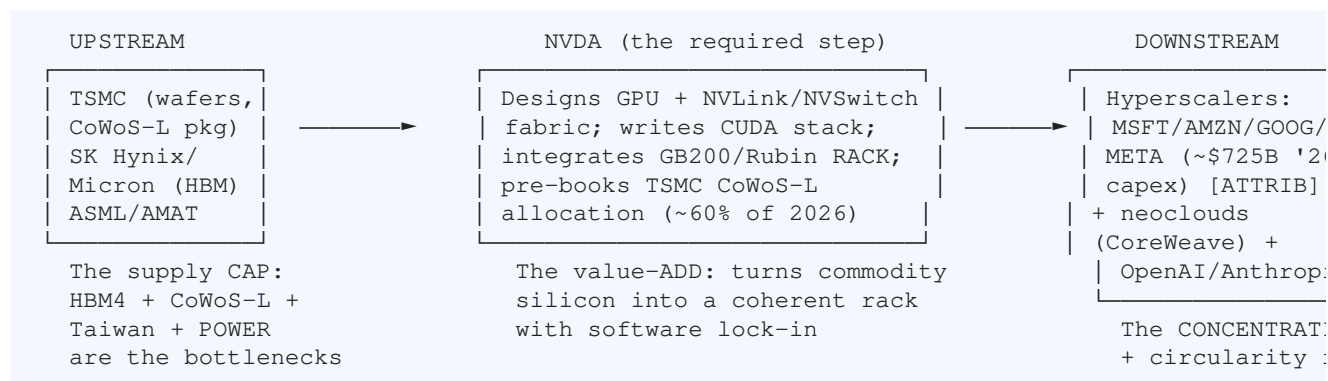
Q (volume) and the dollar pie: - Revenue compounded **26.97B → 60.92B → 130.50B → 215.94B** [FACT] — ~85B of *incremental* revenue in FY26 alone. - Four hyperscalers guiding ~\$725B 2026 capex, **+~77% YoY** [ATTRIBUTED]. - Accelerator TAM ~\$140B (2024) → **>\$440B (2030)** [ATTRIBUTED].

The key PxQ insight: *the dollar pie can expand faster than NVDA's share erodes*. In the sensitivity model, holding share at 75% but moving TAM \$350B→\$550B swings FY29 price **\$194→\$305** [ESTIMATE] — nearly as large as the share lever itself. **Even with share erosion, NVDA's revenue can grow if the market inflects faster.**

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Q4 — Value chain: upstream → NVDA → downstream (Blueprint slides 13–14)



- **Upstream:** TSMC (CoWoS-L), SK Hynix/Micron/Samsung (HBM), ASML/Applied Materials. These — plus **power availability** (RAND: power, not chips, is the binding constraint [ATTRIBUTED]) — *cap deliverable units regardless of design wins*.
- **NVDA's required step:** integration of silicon + interconnect + CUDA + packaging allocation into a rack — the irreplaceable middle.
- **Downstream:** the ~5 hyperscalers, neoclouds, and AI labs. **This is also the risk:** the same ~5 customers that are NVDA's revenue concentration are *funding their own silicon* (TPU/Trainium/Maia/MTIA via AVGO/MRVL) — the customers are the competitors.

Q5 — Financial model, sensitivity matrix, and the dual thesis (Blueprint slides 31–34)

Valuation reconciliation — the 32x vs 43x denominator, honestly

Item	Value	Tag
Spot price	\$209.91	[FACT]
Reported P/E	32.15	[FACT]
Trailing GAAP EPS (FY26)	\$4.90	[FACT]
FY26 net income	\$120.07B	[FACT]
Implied diluted shares (NI ÷ EPS)	~24.5B	[ESTIMATE]
Market cap (spot × 24.5B)	~\$5.14T	[ESTIMATE]
Trailing P/E (209.91 ÷ 4.90)	42.8x	[ESTIMATE]
Implied forward EPS (209.91 ÷ 32.15)	~\$6.53	[ESTIMATE]
Forward/trailing EPS step required	+33%	[ESTIMATE]

Stated plainly: the two numbers are not in conflict — they are the same valuation viewed through two clocks. NVDA trades at **42.8x trailing GAAP** [ESTIMATE on FACT] and **32.15x a forward number** [FACT] that bakes in ~+33% EPS growth to ~\$6.53 [ESTIMATE]. The decisive fact the adversarial work surfaced: **reaching \$6.53 forward EPS at the FY26 net margin of 55.6% [FACT] requires only ~\$286B revenue, i.e. +33% YoY [ESTIMATE]** — roughly *half* the +65.5% NVDA just printed and a fraction of FY25's +114% / FY24's +126% [FACT]. **The forward multiple is built on a below-trend,**

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already-decelerated bar, not a heroic one. PEG is **sub-1.0** [ESTIMATE] vs AMD 180.8x, MRVL 103.4x, AVGO 66.2x [FACT] on slower growth and lower margins.

3-yr (FY2029) model assumptions — every input tagged

Driver	FY2026 anchor	FY2029 base input	Tag
2029 accelerator TAM	~\$140B (2024)	\$440B (mid of ~\$350–550B)	[ATTRIBUTED]
NVDA retained \$ share (the single driver)	~80–90% effective	~ 75% (ASIC inference leakage)	[ASSUMPTION]
Datacenter = % of revenue	—	88%	[ASSUMPTION]
Total revenue	\$215.94B [FACT]	~ \$457B (+40/28/18% path)	[ESTIMATE]
Net margin	55.6% [FACT]	54%	[ASSUMPTION]
Diluted shares	~24.5B [EST]	~ 24.4B	[ASSUMPTION]
Net income	\$120.07B [FACT]	~\$247B	[ESTIMATE]
EPS (FY29)	\$4.90 [FACT]	~ \$10.1	[ESTIMATE]
Exit P/E	32.15x [FACT]	28x (de-rate as growth normalizes)	[ASSUMPTION]

Margin logic: LONG LB2 verification *holds* — **Q4 FY26 GM exited at ~75.0% GAAP** [ATTRIBUTED, NVDA 8-K], so the 71.1% full-year figure is a **ramp-trough average, not a run-rate**. I still haircut to 54% net for prudence against recurring per-architecture (Rubin/Feynman) yield troughs and inference-mix dilution — I do **not** model a return to the 55.8% peak.

The AI-built sensitivity matrix — single technical driver (retained accelerator share) × 2029 TAM

Each cell = **implied FY2029 price / 3-yr IRR from \$209.91** [ESTIMATE]. Held constant: net margin 54% [ASSUMPTION], 24.4B shares [ASSUMPTION], exit 28x [ASSUMPTION], datacenter = 88% of revenue [ASSUMPTION].

Retained accelerator share ↓ / 2029 TAM →			
		(base)	
55% (heavy ASIC encroachment)	\$143 / -12.1%	\$179 / -5.1%	\$224 / +2.2%
65% (meaningful inference leakage)	\$168 / -7.1%	\$212 / +0.3%	\$265 / +8.0%
75% (base — frontier held, inference shared)	\$194 / -2.5%	\$244 / +5.2%	\$305 / +13.3%
85% (moat holds, ASICs stay niche)	\$220 / +1.6%	\$277 / +9.7%	\$346 / +18.1%

Reading: the entire **bottom-left quadrant** (share ≤65% AND TAM ≤\$440B) prices NVDA *below* today's \$209.91 over three years — the short's home turf. The **top-right quadrant** (share ≥75% AND TAM ≥\$440B) delivers low-double-digit IRRs. The base cell (75% / \$440B) = **\$244, +5.2% IRR** — positive but unspectacular: the stock is roughly fairly valued at spot, with the asymmetry sitting in the tails.

Tornado read — what actually moves the value (base cell ±)

1. **Retained share (55%→85%): ~\$98 / 47%** — *the technical driver, dominates.*

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2. **2029 TAM (\$350B→\$550B): ~\$93** (at 75% row, \$194→\$305) — nearly co-equal; the dollar pie can expand faster than share erodes (the LONG's core point).
3. **Exit multiple (22x→34x): ~\$121 spread** — large but a *sentiment* lever, not a fundamental one; weighted least because a de-rate is the likeliest direction.
4. **Margin compression (~46% net) ~\$122 spread** — the *margin short* is the loudest. The bear's whole case rests here, yet it moves value least.

Punchline: the bear is loudest on margin (#4, the *smallest* lever) and the short only truly works if it is *right on share* (#1). Margin compression alone, even to 46% net, still clears ~\$241 at base TAM/multiple — **the margin short is not enough; you need the share short.**

The LONG thesis

"The unit of sale moved from a chip to a coherent rack, and the moat moved with it."

The moat is three stacked, independent layers (software + interconnect + packaging allocation) a competitor must beat *simultaneously*. It monetizes at FY26 **71.1% GM / 60.4% op / 55.6% net** [FACT] — software economics on a hardware BoM, self-funding R&D, near-unlevered. The "margin roll" is a Blackwell-ramp/CoWoS-L trough (Q4 FY26 GM exited ~75% GAAP [ATTRIBUTED]), not structural — and the tornado shows margin is the *smallest* fundamental lever anyway. PxQ redefined upward (\$30K GPU → \$3M rack → ~\$8.8M Rubin rack [ATTRIBUTED]) into a TAM inflecting to >\$440B [ATTRIBUTED]. At 32x forward (sub-1.0 PEG) vs AMD 180.8x / MRVL 103.4x / AVGO 66.2x [FACT], it is the cheapest, highest-margin, fastest-growing name in the group.

- **Valuation [ESTIMATE]:** Base **\$244–283 / +5% to +10% IRR** (share 75%, TAM \$440B, nm 54%, 28x). Bull **\$346–372 / +18% to +21% IRR** (share 85%, TAM \$550B, nm 55%, 30x).
- **Entry / trade [FACT inputs]: accumulate ≤\$195, full size ≤\$180; ~2:1 reward/risk.** Median sell-side ~\$300, 8/8 Buy [FACT].
- **Holding period:** ~3 years (to FY2029).
- **What breaks the long:** an FY23-style air-pocket off a \$5.1T base; net margin sustained below ~50% as inference migrates to ASICs faster than TAM grows; or a CoWoS-L/HBM4/Taiwan supply shock. The asymmetry favors the bull only while GM holds the high-60s and growth stays >25%.

The SHORT thesis

"The rent is already compressing while the multiple still prices perfection."

Not a fraud short — the moat (NVL72 measured up to 28x MI355X on DeepSeek MoE [ATTRIBUTED]) and CUDA are conceded. It is a **margin-and-multiple / share-leakage** short. GM **peaked 75.0% (FY25) and rolled to 71.1% (FY26) during +65.5% growth** [FACT] — margin fell at the *top* of the cycle. Inference (~2/3 of compute [ATTRIBUTED]) migrates off merchant GPUs to custom ASICs at 40–65% lower TCO [ATTRIBUTED]; bears model NVDA inference share falling toward **20–30% by 2028** [ATTRIBUTED]. FY23 is the proven failure mode (rev +0.2%, GM 64.9%→56.9%, op 37.3%→15.7%, EPS \$0.17 [FACT]). Circular financing (>\$40B AI-equity commitments vs \$10.61B cash [FACT]; OpenAI ~\$14B 2026 loss [ATTRIBUTED]) degrades the demand signal. On a **42.8x trailing** name [ESTIMATE], a single soft quarter hits hard.

- **Valuation [ESTIMATE]:** Bear **\$124–142 / -12% to -16% IRR** (share 55–60%, TAM ~\$300–350B, nm 48–50%, 22–24x). Target on the thesis: **\$140–160**.
- **Entry / trigger: tactical, defined-risk — put spread + AVGO/ASIC pair** on a share-loss tell: **GM <67% structural &/or DC rev <+25% &/or frontier-MoE moving off-NVDA.** Not an outright short (8/8 Buy, median ~\$300, deep borrow → squeeze risk).
- **Holding period:** event-driven, ~6–18 months to a confirmed tell.

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- **Why honest:** the adversarial work **refuted (high)** SHORT LB1 (margin "structurally peaked") on all three lenses — Q4 FY26 GM exited ~75% GAAP [ATTRIBUTED]; and the forward EPS bar is *below-trend*, not heroic. The short needs the *share* leg, which only *weakened* the bull — it did not break it.

The senior call (the desk verdict)

LONG-BIASED but two-sided, with entry discipline — accumulate below ~\$195, do not chase at \$210. What tips it long: the adversarial verification was decisive against the SHORT's core claim — **SHORT LB1 (margin "structurally peaked") was refuted high on all three lenses** because Q4 FY26 GM exited ~75% GAAP [ATTRIBUTED], so 71.1% is a ramp-trough average, and the tornado shows margin is the *smallest* fundamental lever anyway. What keeps it two-sided: **SHORT LB2 (inference share leakage) only weakened — it survived** — and the verification surfaced real frontier encroachment (Anthropic→~1M TPUs, Meta TPU-leasing) that dents the LONG's "ASICs can't touch frontier/MoE" claim. The whole call therefore collapses onto one technical number: **NVDA's retained accelerator share.** At ~75% retained and a \$440B 2029 TAM the stock is roughly fairly valued (~+5% IRR at spot), so **the edge is in the entry, not the thesis** — buy the share-holds optionality cheaply below \$195, and flip to a paired underweight only on a confirmed share-loss tell, never on a margin headline alone.

Peer context [FACT]: AMD 542.50 / 180.8x / GM 49.5%; AVGO 397.32 / 66.2x / GM 67.8%; MRVL 301.79 / 103.4x; TSM 469.46 / 39.7x / GM 56.1%; QCOM 228.73 / 24.9x; INTC 139.12 / neg.

Q6 — Investment philosophy (Blueprint slide 35)

Find the one technical variable the whole valuation hangs on, price the tails, and let the entry — not the narrative — carry the edge. Great deep-tech businesses are not won or lost on the loudest debate (here, margins) but on the quietest structural one (here, retained dollar-share as workloads migrate). I underwrite long *and* short on the same fact set, stress the load-bearing claims adversarially until one side *refutes* rather than merely rebuts, and then size to the asymmetry: when a fairly-valued compounder offers free optionality below a disciplined entry, you buy the optionality cheaply rather than chasing the story at spot — and you express the bear case as defined-risk relative value, not a conviction zero, because being *right on the fundamental and wrong on the price for a year* is the most expensive mistake in this sector.

Open questions / what would change the thesis (Blueprint slide 36)

- **Does the frontier stay NVDA-anchored?** Anthropic→~1M TPUs and Meta TPU-leasing [ATTRIBUTED] are the early tell that large-MoE training is *not* CUDA-locked. A second frontier lab going off-NVDA would push the model toward the 65% row.
- **Where does GM settle post-Rubin ramp?** A sustained run-rate below ~67% structural (not a single ramp quarter) would validate the bear's structural read.
- **Does hyperscaler capex ROI prove out?** ~\$725B 2026 capex [ATTRIBUTED] depends on AI-application returns still unproven at that spend; a single capex pause hits a 42.8x name violently.
- **Power, not chips.** If power (RAND: the binding constraint [ATTRIBUTED]) caps deliverable units, share matters less than the TAM ceiling.

Data-quality & source caveats (honest)

- **In-session, no API key.** Reasoning performed by the Claude desk on the deterministic data brief; the LLM agents 04–08 were not run.

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- **Spot \$209.91, reported P/E 32.15, the FY26 anchor are [FACT].** The valuation bridge (**42.8x trailing, ~\$6.53 forward EPS, ~24.5B shares, ~\$5.14T mcap**) is **[ESTIMATE]** derived arithmetically from those facts — not separately reported.
- **The FY2029 model is [ESTIMATE/ASSUMPTION].** TAM (\$350–550B), retained share (the single driver), net margin (54%), share count and exit multiple (28x) are assumptions; the matrix prices/IRRs follow mechanically and are only as good as those inputs.
- **[ATTRIBUTED] items** (Q4 FY26 GM ~75% exit, NVL72 vs MI355X benchmark, ~\$725B capex, Anthropic/Meta TPU moves, Rubin rack ASP ~\$8.8M, ASIC TCO and unit-growth figures, circular-financing details) come from third-party sources cross-referenced in the brief, not audited financials — treat as directional, not precise.
- **Forward EPS is a market-implied figure,** not guidance: ~\$6.53 is simply $209.91 \div 32.15$ **[ESTIMATE]**. The bull's 32x and the bear's 43x are the *same* valuation on two clocks; do not double-count.
- **Two-sided by design.** A "long" mandate still flags the short, and vice versa — the matrix tails, not a single point estimate, are the decision object.